

Hasib Zunair

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WORK EXPERIENCE

Pave

Senior Machine Learning Engineer

Toronto, Canada

May 2025 – Present

- **Delivered >80% automation** for exterior vehicle inspection by designing **multimodal** and **spatiotemporal** DETR-based damage detection system; achieving 92% precision, 87% recall without manual intervention and reducing turnaround time by 85% compared to human inspectors, enabling large-scale automated fleet inspections.
- Architected custom neural network pipelines with 89% accuracy on component identification, automated image quality analysis reducing 85% compute cost via distilling VLMs and offline on-device usage with capture guidance.
- Implemented **multi-task learning** to predict granular damage attributes with a unified model; achieving **>90% precision** reducing manual inspection steps and improving structured metadata for high-fidelity damage profiling.

Decathlon

Machine Learning Engineer

Montreal, Canada

Sept. 2020 – June 2024

- Built real-time basketball video object tracking pipeline; optimized with quantization, pruning, and distillation, cutting memory + latency by 30%; designed **model-assisted labeling workflow**, reducing annotation cost by 85%.
- Implemented **semi-supervised learning** using EfficientNets to **train on unlabeled data**, resulting 10% accuracy, 5× lower cloud usage; developed **generative AI models** using GANs, pix2pixHD, OpenPose, improving realism by 20% and fine-tuned YOLO for bike brand detection from marathon images, driving faster, cheaper market insights.
- **Tools:** Python, PyTorch, OpenCV, ONNX, CoreML, FastAPI, Docker, GCP.

Concordia University

Graduate Machine Learning Researcher

Montreal, Canada

Sept. 2019 – Dec. 2024

- Led publications and presented at top-tier conferences (**WACV**, **BMVC**), oral presentation at **BMVC'22** (top 5% acceptance rate), journals (**IEEE TMI**, IF: 10.6) and workshops (**CVPR**, **ICML**), resulting in **1600+ citations**.
- Designed and implemented novel 2D/3D deep learning algorithms, outperforming state-of-the-art in accuracy, data efficiency, and compute; tailored vision models like ConvNet, ViT, DINO, YOLO, CycleGAN using unsupervised, self-supervised, and zero-shot learning for image recognition, generation, detection, and segmentation.
- Mentored 15 students from undergraduate to Ph.D. level through research projects, publishing at several journals.

Ericsson

Machine Learning Specialist

Montreal, Canada

Oct. 2021 – Mar. 2022 & Feb. 2024 – June 2024

- Led hands-on training for **GPT-style LLMs**, **fine-tuning** and **data science tooling**; mentored via pair programming, code reviews, and experiment design to drive successful end-to-end project delivery.
- Guided 21 Ericsson staff in building seq2seq models (Transformers, LSTM autoencoders) for time-series forecasting and anomaly detection using PyCaret, PyOD and Darts, improving model performance and production readiness.

SKILLS

- **Programming Languages:** Python, C++.
- **Tools & Libraries:** PyTorch, Unsloth, TensorFlow/Keras, scikit-learn, OpenCV, ONNX, CoreML.
- **Cloud Infrastructure and MLOps:** GCP, FastAPI, Docker, Gradio, GitHub Actions, Kubernetes, RunPod.

PROJECTS & PUBLICATIONS & OPEN-SOURCE CONTRIBUTIONS

- **Multi-Modal AI (2025):** Created domain-specific datasets to enable two systems: **Afiyah**, a multimodal OCR pipeline using LLM fine-tuning and RAG-style retrieval to extract and explain harmful ingredients. **FitLensAI**, a Llama 3-based vision-language model fine-tuned with LoRA for multilingual, multi-turn fitness conversations from workout images.
- **Unsupervised Object Localization at BMVC (2024):** Created **PEEKABOO** using DINO to segment unfamiliar objects without any additional training, that is accurate, faster and lighter than competing state-of-the-art methods.
- **AICITY competition at CVPR (2022):** Led a team for **VISTA**, by training Vision Transformers (ViTs) on synthetic data, to recognize and count products in videos for retail checkout to improve operational efficiency, achieving 3rd place.
- **Contributions to core ML/CV libraries with >75K GitHub stars:** Added Python code to Kornia (**MS-SSIM loss**), TensorFlow (**3D image classification**), and YOLOv6 (**ONNX export fix**), driving higher accessibility and usability.

EDUCATION

- **Concordia University** Doctor of Philosophy (Ph.D.) and M.A.Sc in Machine Learning & Artificial Intelligence
- **North South University** B.Sc. in Electrical & Computer Engineering